

Mahesh Babu Vishnumolakala

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SUMMARY

Final-year CSE student (8.89 CGPA) with a published, peer-reviewed IEEE paper on transformer-based language identification. Built and shipped ML systems end to end, from fraud detection to retrieval-augmented search, and contributed to a live production product as an AI engineering intern.

TECHNICAL SKILLS

ML, DL: PyTorch, HuggingFace Transformers, Scikit-learn, Fine-Tuning, Feature Engineering, Model Evaluation, MLflow

Agentic AI / LLM: RAG Pipelines, Vector Databases, Agentic Patterns, Prompt Engineering

Languages & Data: Python, Java, SQL

Backend, Infra & Tools: FastAPI, REST APIs, Docker, AWS, Git

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems

EXPERIENCE

AI Engineering Intern | Perseverance AI (Lensy) | perseveranceai.com Sep 2026 – Present

Contributing to an AI-readiness auditing product for technical documentation

- Built a fallback pipeline for JS-rendered pages, adding retry handling for rate-limit failures and a consent step before analysis.
- Found and fixed classification rules that were wrongly rejecting valid URLs, improving pipeline reliability.
- Rebuilt the product frontend from Figma design to live deployment, restructuring all routes without breaking existing logic.

Research Assistant | Hinglish Language ID | [IEEE Xplore](https://www.ieee.org/xplore) Dec 2025 – Apr 2026

Dept. of CSE, LPU, under Prof. Enjula Uchoi. Presented at ICICI 2026.

- Fine-tuned and compared four transformer models (mDeBERTa-v3, XLM-RoBERTa, mBERT, DistilBERT) on 74,813 labeled tokens; best model reached 96.68% accuracy and 0.88 macro-F1.
- Paper published on IEEE Xplore and indexed on Google Scholar; findings presented at ICICI 2026.

PROJECTS

Fraud Detection Model Benchmarking (IEEE-CIS Dataset) | [GitHub](https://github.com) Aug 2026

- Trained and compared XGBoost, LightGBM, and a custom PyTorch MLP on over 590K transactions, evaluating all three models under the same conditions to make the comparison fair.
- Engineered 15+ features, applied SMOTE to training split only to prevent data leakage.
- LightGBM came out on top with a 0.961 AUC and 0.72 F1 score. Tuned the decision threshold afterward and pushed fraud recall from 0.59 up to 0.69.
- Tracked experiments and registered the winning model in the MLflow Model Registry.

Semantic Code Search Engine | [GitHub](https://github.com) | [Live Demo](#) Apr 2026

- Built a RAG-based search backend in FastAPI that takes a plain-English query and finds matching code, using transformer embeddings and ChromaDB for retrieval.
- Added a cross-encoder reranking step to clean up the initial results, then stress-tested it with tricky queries to find where retrieval broke down.
- Containerized the whole thing with Docker and deployed it on HuggingFace Spaces and Vercel, with a React frontend on top.

CERTIFICATIONS

McKinsey Forward Program | [McKinsey & Company](#) Jun 2026

OCI Generative AI Professional | [Oracle University](#) Oct 2025

ACHIEVEMENTS

Secured 2nd place in a university-level hackathon for a malicious URL detection system built with ML Feb 2026

Ranked in the top 10% of the university Mar 2025

EDUCATION

Lovely Professional University Phagwara, Punjab

Bachelor of Technology, Computer Science and Engineering | CGPA: 8.89

Aug 2023 – Present